

PROJECT AND TEAM INFORMATION

Project Title

E-Commerce Behaviour & Recommendation System

Student / Team Information

Team Name: Alone Warrior

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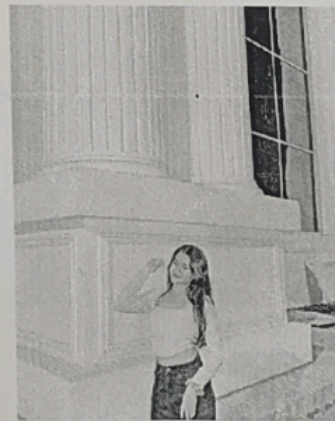
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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

Nowadays, people use e-commerce websites for almost everything, from buying clothes and electronics to groceries and other daily-use products. Because of this, these platforms generate a huge amount of data about what customers search for, view, add to their cart, and purchase.

Handling such a large amount of data manually is difficult. Also, different customers have different interests, so showing the same products to everyone may not give a good shopping experience. This motivated us to develop a system that can study customer behavior and understand their preferences using Machine Learning.

The main aim of this project is to use a large e-commerce dataset to analyze customer activities and provide personalized product recommendations. The system can also help identify different types of customers and predict their possible purchasing behavior. This can make the shopping experience more relevant for customers while helping businesses understand their customers better.

State of the Art / Current solution (1 pt)

Today, most e-commerce websites use recommendation systems to help customers find products they may be interested in. These systems usually look at things like previous purchases, products viewed, searches, ratings, and items added to the cart. Based on this information, they suggest similar or related products to the customer. Common methods used for this are collaborative filtering and content-based filtering. Some platforms also combine both methods to get better recommendations. These systems can work well when enough information about the customer is available.

However, existing systems can have problems with new users who have little or no purchase history. Managing and processing a very large amount of customer activity data is another challenge.

Simple recommendation systems may also focus mainly on suggesting products without giving much information about overall customer behavior. Our proposed system aims to overcome some of these limitations by using a large e-commerce dataset and Machine Learning.

It will study customer activities, identify buying patterns, and provide personalized product recommendations along with useful customer behavior insights.

Project Goals and Milestones (2 pts)

Project Goals:

- To collect and work with a large e-commerce customer behavior dataset.
- To understand customer activities such as product views, searches, cart actions, and purchases.
- To identify different types of customers based on their shopping behavior.
- To develop a Machine Learning based recommendation system that suggests relevant products.
- To predict customer purchase behavior using suitable ML algorithms.
- To store and manage customer and product data using a database.
- To create a simple and user-friendly web interface for showing recommendations and customer insights.
- To evaluate the performance of the developed models and improve their accuracy.

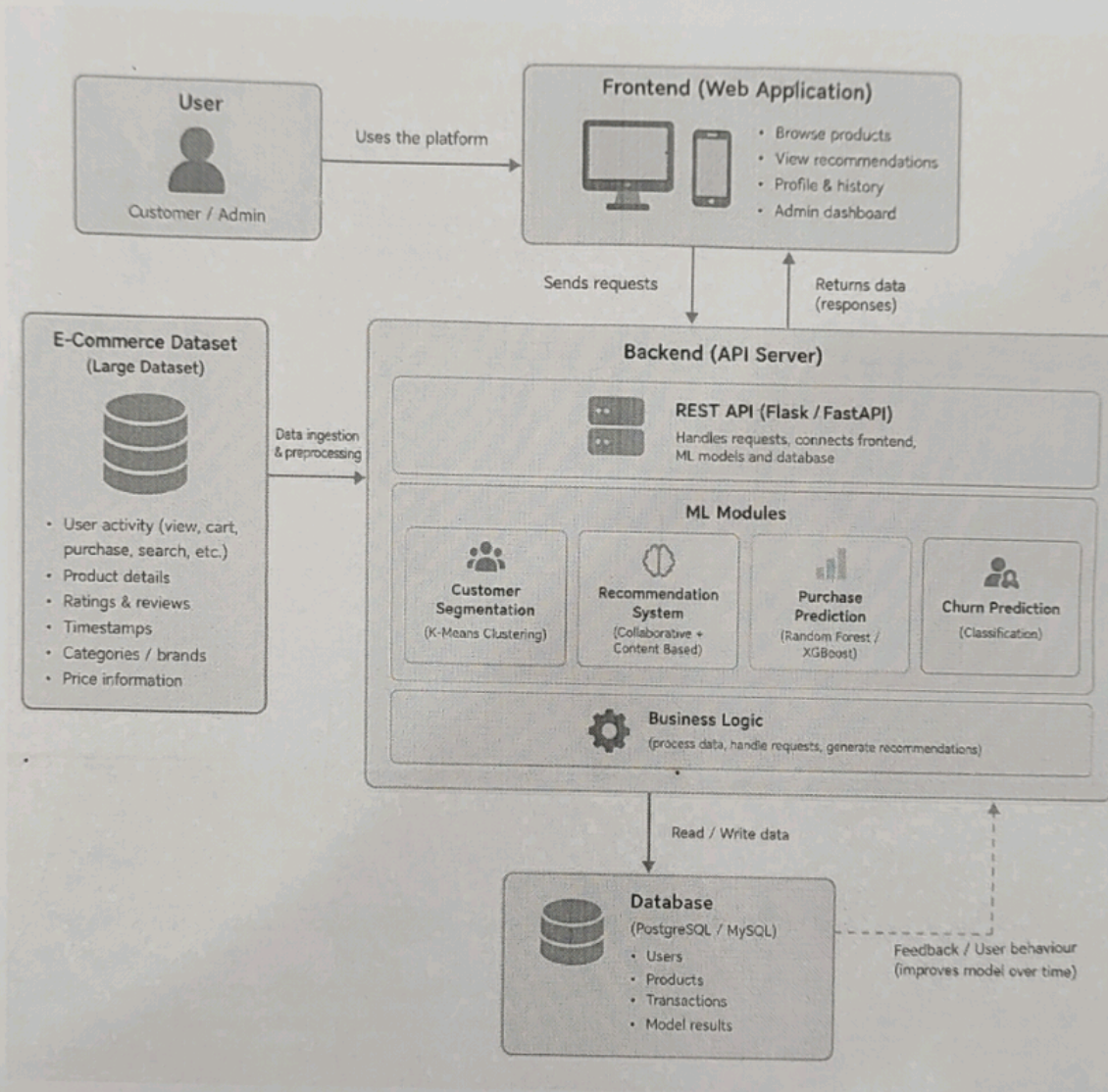
Project Milestones:

1. Requirement Analysis – Finalize the project requirements and features.
2. Data Collection – Find and prepare a large e-commerce dataset.
3. Data Preprocessing – Clean the data and handle missing or duplicate records.
4. Data Analysis – Study customer behavior using EDA and visualizations.
5. Model Development – Train and test ML models for segmentation, prediction, and recommendations.
6. Backend & Database – Develop APIs and connect the system with the database.
7. Frontend Development – Create the user interface and recommendation dashboard.
8. Testing & Evaluation – Test the complete system and measure model performance.
9. Final Deployment & Documentation – Integrate all components and prepare the final report and presentation.

Project Approach (3 pts)

The project will be developed step by step, starting with collecting a large e-commerce customer behavior dataset. The dataset will contain information such as customer activity, product details, views, searches, cart actions, purchases, and timestamps. First, the collected data will be cleaned and prepared by handling missing values, duplicate records, and unnecessary information. After that, Exploratory Data Analysis (EDA) will be performed to understand customer behavior and find useful patterns in the data. Next, different Machine Learning techniques will be applied. Customer segmentation will be done to group users with similar shopping behavior, while recommendation techniques will be used to suggest products based on customer preferences and past activities. A separate model can also be used to predict the possibility of a customer purchasing a product. The processed data and results will be stored in a database. A backend will connect the ML models with the application, and a simple frontend will be developed where users can view products and personalized recommendations. Finally, the complete system will be tested using suitable performance measures and improved based on the results.

System Architecture (High Level Diagram)(2 pts)



Project Outcome / Deliverables (1 pts)

The main outcome of this project will be a working e-commerce system that can analyze customer behavior and provide personalized product recommendations. The system will be developed using a large customer activity dataset and Machine Learning techniques.

The expected deliverables are

- A cleaned and processed large e-commerce dataset.
- EDA reports and visualizations showing customer and product behavior.
- A Machine Learning model for customer segmentation.
- A recommendation model for suggesting relevant products to users.
- A model for predicting customer purchase behavior.
- A database for storing user, product, activity, and transaction data.
- A backend API connecting the database and ML models with the application.
- A simple web-based frontend for browsing products and viewing personalized recommendations
- A dashboard showing useful customer behavior and recommendation insights
- Final testing and evaluation of the developed models and complete system.

Assumptions

It is assumed that the available e-commerce dataset contains enough customer and product information to identify useful shopping patterns.

The user activity data is assumed to be reasonably accurate and suitable for Machine Learning. It is also assumed that users will interact with the system by viewing, searching, adding, or purchasing products, which will provide data for making recommendations.

The project assumes that the selected Machine Learning algorithms will be able to find meaningful patterns in the large dataset. The database and web application are also assumed to have enough resources to handle the processed data and provide results within a reasonable time

References

1. Kaggle – E-commerce datasets
2. Scikit-learn Documentation – Machine Learning
3. Pandas Documentation – Data Analysis
4. Python Documentation – Programming
5. PostgreSQL Documentation – Database Management